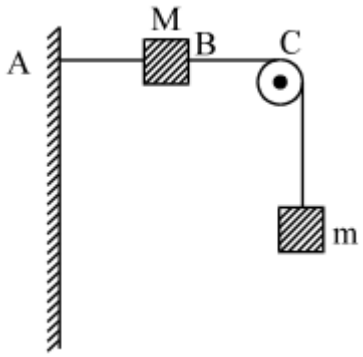
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A string with one end fixed on a rigid wall passing over a fixed frictionless pulley at a distance of $2\ \mathrm{m}$ from the wall has a point mass $\mathrm{M} = 2\ \mathrm{kg}$ attached to it at a distance of $1\ \mathrm{m}$ from the wall.
A mass $\mathrm{m} = 0.5\ \mathrm{kg}$ attached at the free end is held at rest so that the string is horizontal between the wall and the pulley and vertical beyond the pulley. What will be the speed with which the mass M will hit the wall when mass m is released. $\left(\mathrm{g} = 9.8\ \mathrm{m/s^2}\right)$



(A) $3.3\ \mathrm{m/s}$

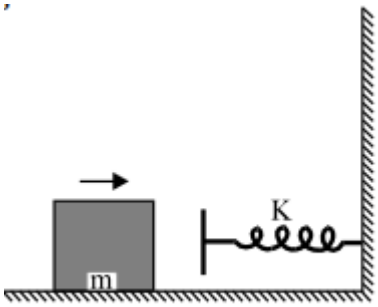
(B) $1.2\ \mathrm{m/s}$

(C) $6.4\ \mathrm{m/s}$

(D) $8\ \mathrm{m/s}$

Q.2

$1.0\ \mathrm{kg}$ block collides with a horizontal weightless spring of force constant $2.0\ \mathrm{N/m}$. The block compresses the spring $4.0\ \mathrm{m}$ from the rest position. Assuming that the coefficient of kinetic friction between the block and the horizontal surface is 0.25. what was the speed of the block at the instant of collision ?



(A) $7.18\ \mathrm{m/s}$

(B) $17.18\ \mathrm{m/s}$

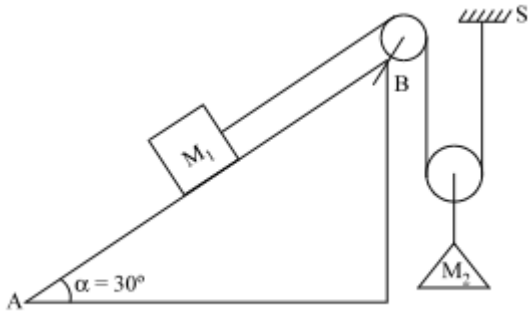
(C) $3.18\ \mathrm{m/s}$

(D) $6.18\ \mathrm{m/s}$

@IITJEE_Advanced

Q.3

A $30\ \mathrm{kg}$ block M_1 rests on a smooth inclined plane AB. If is attached by a flexible inextensible string over frictionless light pulleys to a $45\ \mathrm{kg}$ mass M_2 and a support S. If the angle of inclination of the plane is 30°, find in what distance will the block M_1 attain the speed of $3\ \mathrm{m/s}$?



(A) $1.525\ \mathrm{m}$

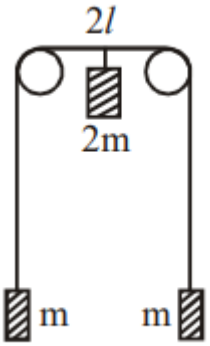
(B) $2.525\ \mathrm{m}$

(C) $4.525\ \mathrm{m}$

(D) $5.525\ \mathrm{m}$

Q.4

Equal weights of mass m are suspended from the ends of a very long string thrown over two small stationary pulleys at a distance of 2ℓ from each other (Fig.). Find the velocity of the weights after a sufficiently long time if a weight with a mass $2\ \mathrm{m}$ is attached to the middle of the string.



(A) $\sqrt{g\ell}$

(B) $\sqrt{2\ g\ell}$

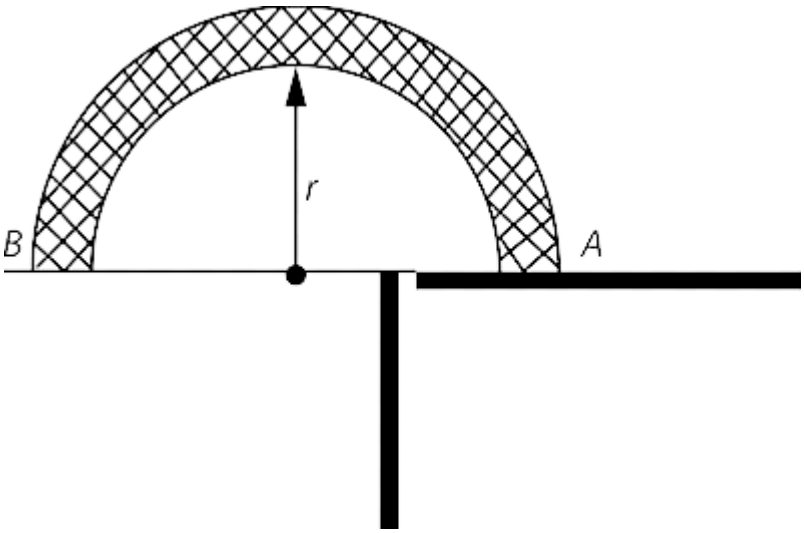
(C) $\sqrt{3\ g\ell}$

(D) $\sqrt{5\ g\ell}$

Q.5

A smooth semicircular tube AB of radius R is fixed in a verticle plane and contains a heavy flexible chain of length πR. Find the velocity v with which it will emerge from the open end ' B ' of tube, when slightly displaced.

@IITJEE_Advanced



(A) $\sqrt{2gR(2\pi + 2/\pi)}$

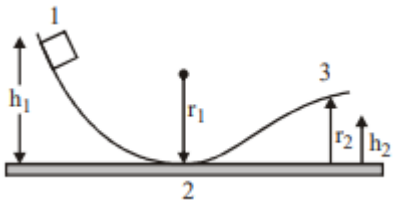
(B) $\sqrt{\frac{gR}{2}\left(\frac{\pi}{4} + 4\pi\right)}$

(C) $\sqrt{2gR\left(\frac{2}{\pi} + \frac{\pi}{2}\right)}$

(D) $\sqrt{gR\left(\pi + \frac{1}{\pi}\right)}$

Q.6

A block of mass m is released from a height h_1 along a smooth track as shown in figure.



Determine the minimum safe value of radius of curvature at point 3 , so that the block does not fly off the track -

(A) $(h_1 - h_2)$

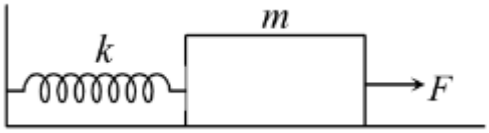
(B) $2\,(\,h_1 - h_2)$

(C) $\frac{(h_1 - h_2)}{3}$

(D) $\frac{2(\,h_1 - h_2)}{3}$

Q.7

A block of mass ' m ' is pulled by horizontal constant force $F = 5\mu mg$ over a rough surface of coefficient of friction μ as shown. Initially spring was at its natural length, the position where block will finally comes to rest will be



@IITJEE_Advanced

(A) $\frac{\mu mg}{k}$

(B) $\frac{4\mu mg}{k}$

(C) $\frac{6\mu mg}{k}$

(D) $\frac{8\mu mg}{k}$

Q.8

A wind - powered generator converts wind energy into electric energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to

(A) v

(B) v^2

(C) v^3

(D) v^4

Q.9

A block is suspended by an ideal spring of force constant k. If the block is pulled down by applying a constant force F and if maximum displacement of the block from its initial position of rest is δ, then -

(A) $\frac{\mathrm{F}}{\mathrm{k}} < \delta > \frac{2\,\mathrm{F}}{\mathrm{k}}$

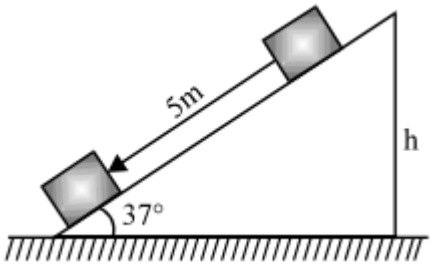
(B) $\delta = \frac{2\,\mathrm{F}}{\mathrm{k}}$

(C) work done by force F is equal to $\mathrm{F}\delta$

(D) increase in energy stored in the spring is $\mathrm{K}\delta^2$

Q.10

A block is placed on the top of a plane inclined at 37° with horizontal. The length of the plane is $5\ \mathrm{m}$. The block slides down the plane and reaches the bottom. Choose the correct options.



(A) The speed of the block at the bottom is $7.67\ \mathrm{m/s}$ (if the inclined plane is smooth).

(B) The speed of the block at the bottom is $6.26\ \mathrm{m/s}$ (if the coefficient of friction is 0.25).

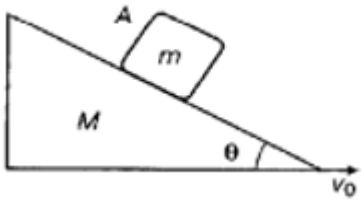
(C) The speed of the block at the bottom is $6.67\ \mathrm{m/s}$ (if the inclined plane is smooth).

(D) The speed of the block at the bottom is $5.26\ \mathrm{m/s}$ (if the coefficient of friction is 0.25).

Q.11

@IITJEE_Advanced

The triarygular block is moving horizontally with constant velocity V_0 as shown in the figure. The block A always remain stationary with respect to triangular block. In time t,



(A) The work done by normal reaction on triangular block is $W_1 = \frac{-mg}{2} v_t t \sin(2\theta)$

(B) The work done by normal reaction on triangular block is zero

(C) The work done by normal reaction on block A is $W_2 = \frac{mg}{2} v_0 t \sin(2\theta)$

(D) The work done by normal reaction (between block A and the triangular block) is zero on the system (triangular block $+$ block A) in any frame of reference

Q.12

Passage:
A single conservative force $\mathrm{F(x)}$ acts on a $1.0\ \mathrm{kg}$ particle that moves along the x-axis. The potential energy $U(x)$ is given by $U(x) = 20 + (x-2)^2$ where x is in meters. At $x = 5.0\ \mathrm{m}$ the particle has a kinetic energy of $20\ \mathrm{J}$.
Question: What is the mechanical energy of the system -

(A) $35\ \mathrm{J}$

(B) $64\ \mathrm{J}$

(C) $86\ \mathrm{J}$

(D) $49\ \mathrm{J}$

Q.13

Passage:
A single conservative force $\mathrm{F(x)}$ acts on a $1.0\ \mathrm{kg}$ particle that moves along the x-axis. The potential energy $U(x)$ is given by $U(x) = 20 + (x-2)^2$ where x is in meters. At $x = 5.0\ \mathrm{m}$ the particle has a kinetic energy of $20\ \mathrm{J}$.
Question: The maximum and minimum values of x respectively are $-$

(A) $7.38\ \mathrm{m}, -3.38\ \mathrm{m}$

(B) $6.38\ \mathrm{m}, -4.38\ \mathrm{m}$

(C) $7.38\ \mathrm{m}, -2.38\ \mathrm{m}$

(D) $6.38\ \mathrm{m}, -2.38\ \mathrm{m}$

Q.14

Passage:
A single conservative force $\mathrm{F(x)}$ acts on a $1.0\ \mathrm{kg}$ particle that moves along the x-axis. The potential energy $U(x)$ is given by $U(x) = 20 + (x-2)^2$ where x is in meters. At $x = 5.0\ \mathrm{m}$ the particle has a kinetic energy of

@IITJEE_Advanced

20 J.

Question: The maximum kinetic energy of the particle and the value of x at which maximum kinetic energy occurs -

(A) 29J, 0 m

(B) 49 J, 0 m

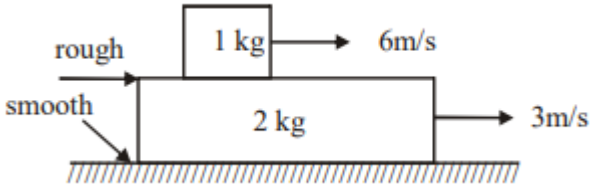
(C) 49 J, 2 m

(D) 29 J, 2 m

Q.15

Passage:

In the figure shown, upper block is given a velocity 6 m/s and lower block a velocity 3 m/s. When relative motion between them stops (Here block 2 kg is very long). Treat all quantities in ground frame.



Question: Choose the correct options -

(A) Work done by friction on 1 kg block is negative

(B) Work done by friction on 2 kg block is positive

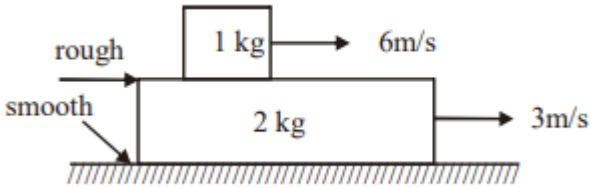
(C) Both (A) and (B) are correct

(D) (A) is correct but (B) is incorrect

Q.16

Passage:

In the figure shown, upper block is given a velocity 6 m/s and lower block a velocity 3 m/s. When relative motion between them stops (Here block 2 kg is very long). Treat all quantities in ground frame.



Question: Net work done by friction -

(A) -3 J

(B) -10 J

(C) -7 J

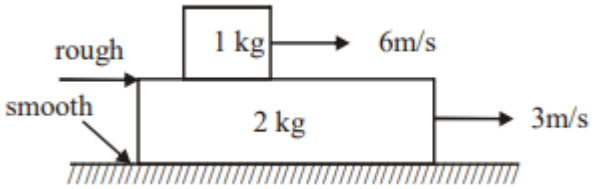
(D) -4 J

@IITJEE_Advanced

Q.17

Passage:

In the figure shown, upper block is given a velocity $6\ \mathrm{m/s}$ and lower block a velocity $3\ \mathrm{m/s}$. When relative motion between them stops (Here block $2\ \mathrm{kg}$ is very long). Treat all quantities in ground frame.



Question: Loss in K.E. of system (2kg + $1\ \mathrm{kg}$ block)

(A) $3\ \mathrm{J}$

(B) $10\ \mathrm{J}$

(C) $7\ \mathrm{J}$

(D) $4\ \mathrm{J}$

Q.18

A man pushes a block of $30\ \mathrm{kg}$ along a level floor at a constant speed with a force directed at 45° below the horizontal. If the coefficient of friction is 0.20, then match the column -

Column I	**Column II**
(A) Work done by all forces exerted by the surface on the block in 20m.	(p) zero
(B) Work done by the force of gravity	(q) 1500 J
(C) Work done by the man on the block in pushing it through 10m	(r) 750 J
(D) Net force on the block	(s) 30 J

(A) A-(q) B-(p) C-(s) D-(r)

(B) A-(p) B-(r) C-(s) D-(q)

(C) A-(r) B-(p) C-(s) D-(q)

(D) A-(q) B-(p) C-(r) D-(p)

Q.19

A block of mass $m = 1$ kg moving on a horizontal surface with speed $u = 2\ \mathrm{m\ s^{-1}}$ enters a rough horizontal patch ranging from $x = 0.10$ m to $x = 2.00$ m. If the retarding force f_r on the block in this range is inversely proportional to x over this range i.e.

$f_r = -\frac{k}{x}$ for $0.10 < x < 2.00$

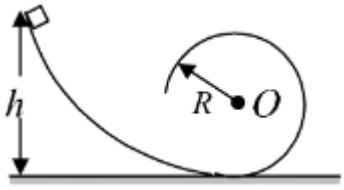
$f_r = 0$ for $x < 0.10$ and $x > 2.00$

If $k = 0.5$ J then, the speed of this block (in $\mathrm{m\ s^{-1}}$) as it crosses the patch is (use $\ln(20) = 3$)

@IITJEE_Advanced

Q.20

A particle is released from rest from the top of an arc, which later curves into a circular track of radius ' R ' as shown. If the minimum height ' h ' from where it should be released so that it is able to complete the circle is kR, then find k.



Answers & Solutions

Q.1 Answer:

—

Solution:

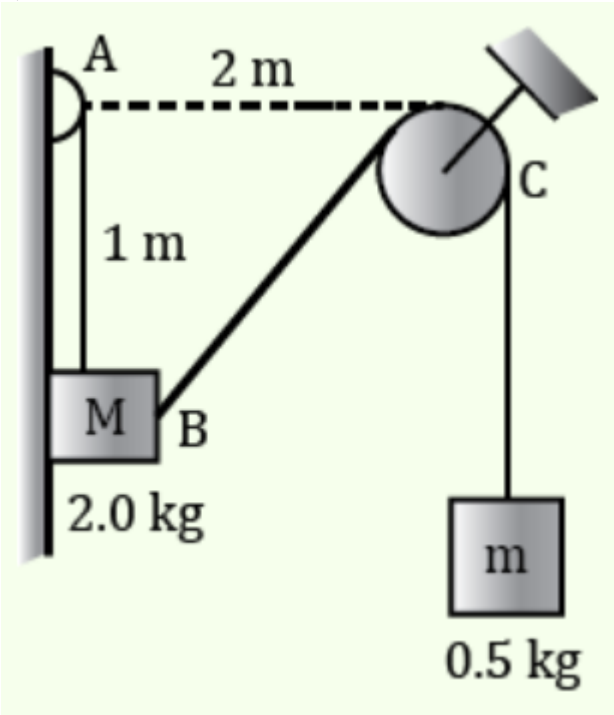
Let the mass M strikes the wall at D, so that string supporting the mass m is pulled up by a height h given by $\mathrm{h} = \mathrm{DC} - \mathrm{BC} = \sqrt{\mathrm{AD}^2 + \mathrm{AC}^2} - \mathrm{AD} \quad [\text{as } \mathrm{AD} = \mathrm{AB} = \mathrm{BC} = 1\ \mathrm{m}]$,
i.e., $\mathrm{h} = \sqrt{1^2 + 2^2} - 1 = \sqrt{5} - 1 = 1.236\ \mathrm{m}$

Now as the string is inextensible the speed of both the masses will be equal at all times (but will be different at different time).

If V **is the speed when** M **hits the wall, gain in K.E. of the system will be** $\frac{1}{2}(\mathrm{M}+\mathrm{m})\mathrm{V}^2$**, gain in P.E. mgh and loss in P.E.** $\mathrm{Mg} \times \mathrm{AD} = \mathrm{Mg}$ **(as** $\mathrm{AD} = 1\ \mathrm{m}$**).**

So by conservation of mechanical energy : $\frac{1}{2}(\mathrm{M}+\mathrm{m})\mathrm{V}^2 + \mathrm{mgh} = \mathrm{Mg} \times 1$

(Gain in Mechanical energy) $=$ **(loss in ME)** **i.e.,** $\frac{1}{2}(2+0.5)\mathrm{V}^2 = 9.8(2 - 0.5 \times 1.236) \Rightarrow \mathrm{V} = \sqrt{\frac{9.8 \times 1.382}{1.25}} \cong 3.3\ \mathrm{m/s}$



Q.2 Answer:

$7.18\ \mathrm{m/s}$

@IITJEE_Advanced

Solution:

Let v the required speed at the beginning of the collision with the spring.

Then according to law of conservation of energy.

Initial K.E. of block = (elastic energy stored in the spring) = (energy spent in doing work against friction)

i.e. $\frac{1}{2}mv^2 = \frac{1}{2}Kx^2 + \mu_k mgx$

Here $m = 1.0\ kg\ K = 2.0\ N/m.x = 4.0, \quad \mu_k = 0.25$

$\therefore \quad \frac{1}{2} \times 1.0 \times v^2 = \frac{1}{2} \times 2.0 \times (4.0)^2 + 0.25 \times 1.0 \times 9.8 \times 4.0$

i.e. $\frac{1}{2}v^2 = 16 + 9.8 = 25.8 \quad$ or $\quad v^2 = 2 \times 25.8 = 51.6$

This gives $v = 7.18\ m/s$

Q.3 Answer:

$2.525\ m$

Solution:

Method 1 : Suppose the block M_1 moves up a distance x, the block M_2 moves down through a distance $x/2$ only.

Hence if v be the velocity of M_1 at any instant the velocity of M_2 will be $v/2$ at that instant.

The initial kinetic energy of the system (K.E.) $_i = 0$

When the block M_1 has a velocity $3\ m/s$. the K.E., of the system is

$(K.E.)_f = \frac{1}{2}(30)(3)^2 + \frac{1}{2}(45)\left(\frac{3}{2}\right)^2\ J = 135 + 50.625 = 185.6\ J$

The gain in K.E. is equal to the loss of P.E. of the system when M_1 rises up by a distance along AB and M_2 descends down by a height $x/2$

$\therefore \quad$ loss in P.E. $= (45\ g)\left(\frac{x}{2}\right) - (30\ g \sin 30^\circ)(x) = \frac{x}{2}(45\ g - 30\ g) = x15\ g$

Now, $\frac{15\ g}{2} \cdot x = 185.6\ J \quad$ or $\quad x = \frac{185.6 \times 2}{15 \times 9.8} = 2.525\ m$

The block M_1 has to move up the plane by a distance $x = 2.525\ m$ so that it will attain a velocity of $3\ m/s$

Method 2 : Application of K.E. equation in this case yields

$\frac{1}{2}(M_1 + M_2)\,u^2 - (M_1\ g \sin)\,x + \frac{x}{2}M_2\ g = \frac{1}{2}M_1 v^2 + M_2\left(\frac{v}{2}\right)^2$

Here $u = 0, v = 3\ m/s : x = 2.525\ m$

Q.4 Answer:

$\sqrt{g\ell}$

Solution:

Assume that the weight $2\ m$ lowers through a distance of H.

The weight m will accordingly rise to the height h on the basis of the law of conservation of energy,

$2mgh + \frac{2mv^2{}_1}{2} + \frac{2mv_2^2}{2} = 2mgH$, or $v^2{}_1 + v^2{}_2 = 2\ g(H - h)$, where v_1 is the velocity of the weights m and v_2 that of the weight $2m$.

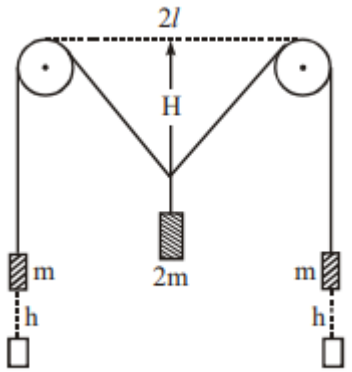
When the weight $2\ m$ lowers its velocity v_2 approaches the velocity v_1 since the angles between the parts of the string thrown over the pulleys tend to zero.

In the limit, $v_2 \cong v_1$.

At the same time, $H - h \cong \ell (H >> h)$

@IITJEE_Advanced

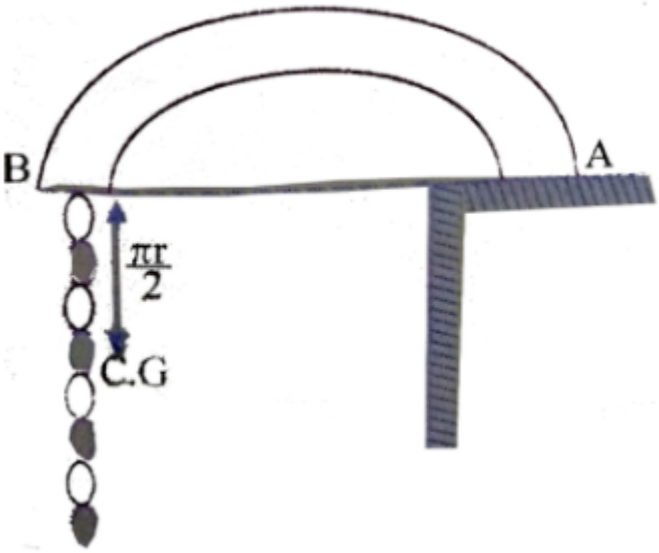
Therefore, the maximum velocity of the weights is $v = \sqrt{g\ell}$



Q.5 Answer:

$\sqrt{2gR\left(\frac{2}{\pi}+\frac{\pi}{2}\right)}$

Solution:



Centre of fravity if s semicircular arc is at a distance $\frac{2r}{\pi}$ from the centre.

Initial potential energy $U_i = (\lambda\pi r)g\left(\frac{2r}{\pi}\right)$

Final potential energy $U_f = (\lambda\pi r)g\left(\frac{-\pi r}{2}\right)$

When the chain is completely slipped off the tube, all the links of the chain have the seme velocity v.

kinetic energy of chain $k = \frac{1}{2}mv^2 = \frac{1}{2}(\lambda\pi r)v^2$

From conservation of energy,

$\lambda\pi rg\left(\frac{2r}{\pi}\right) = (\lambda\pi r)g\left(\frac{-\pi r}{2}\right) + \frac{1}{2}(\lambda\pi r)v^2$

On solving we get, $v = \sqrt{2rg\left(\frac{2}{\pi}+\frac{\pi}{2}\right)}$

@IITJEE_Advanced

Q.6 Answer:

$2\left(\mathrm{h}_1-\mathrm{h}_2\right)$

Solution:

(i) Velocity at point $2:\mathrm{v}_2=\sqrt{2 \mathrm{gh}_1}$

Force on block at $2:\mathrm{F}_2=\mathrm{mg}+\frac{\mathrm{mv}_2^2}{\mathrm{r}_1}=\mathrm{mg}+\frac{\mathrm{m} 2 \mathrm{gh}_1}{\mathrm{r}_1}$

(ii) Velocity at point $3: v_3=\sqrt{2 g\left(h_1-h_2\right)}$

If block does not fly : $\mathrm{v}_3 \leq \sqrt{\mathrm{gr}_2}$

$\Rightarrow 2 \mathrm{~g}\left(\mathrm{~h}_1-\mathrm{h}_2\right) \leq \mathrm{gr}_2 \Rightarrow \mathrm{r}_2 \geq 2\left(\mathrm{~h}_1-\mathrm{h}_2\right)$

Q.7 Answer:

$\frac{4 \mu m g}{k}$

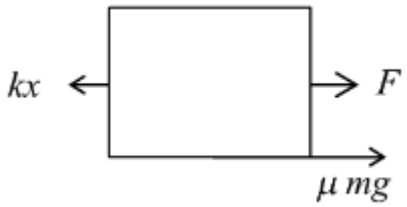
Solution:

Block will return after maximum elongation.

i.e. $F \cdot \mathrm{x}_{\max }-1 / 2 K x_{\max }^2-\mu m g x_{\max }=0$

$x_{\max }=\frac{2(F-\mu m g)}{k}=\frac{8 \mu m g}{k}$

So block will finally comes to rest while returning i.e. $v=0 \ \& \ a=0$



By work energy theorem while returning

$-\left(\frac{1}{2} k x^2-\frac{1}{2} k x_{\max }^2\right)-(F+\mu m g)\left(x_{\max }-x\right)=0 \Rightarrow, x=\frac{4 \mu m g}{k}$

$\therefore$ **(b)**

Q.8 Answer:

v^3

Solution:

Power $=\vec{F} \cdot \vec{v}=F v$

$F=v\left(\frac{d m}{d t}\right)$

$=v\left\{\frac{d(\rho \times(\text { volume }))}{d t}\right\} \quad (\rho=\textbf{density})$

$=\rho v\left\{\frac{d(\text { volume })}{d t}\right\}=\rho v(A v)=\rho A v^2$

Power $P=\rho A v^3$

Or $P \propto v^3$

Q.9 Answer:

$\delta=\frac{2 \mathrm{~F}}{\mathrm{k}}$

Solution:

@IITJEE_Advanced

Let mass of the block hanging from the spring be m.

Then initial elongation of the spring will be equal to $\mathrm{mg/k}$.

When the force F applied to pull the block down, the work done by F and further loss of gravitational potential energy of the block is used increase strain energy of this spring.

$(F\delta = mg\delta) = \left[\frac{1}{2}k\left(\frac{mg}{k}+\delta\right)^2 - \frac{m^2g^2}{2k}\right]$

From this eq., $\delta = \frac{2F}{\mathrm{k}}$. Hence option (B) is correct.

Since, a constant force is applied on the block to pull it down, therefore during this process, work done by force, $\mathrm{W} = \mathrm{F}\delta$.

Hence, option (C) is also correct.

Increase in energy of the spring is equal to $\left[\frac{1}{2}\mathrm{k}\left(\frac{\mathrm{mg}}{\mathrm{k}}+\delta\right)^2 - \frac{\mathrm{m}^2\,\mathrm{g}^2}{2\mathrm{k}}\right]$

From this equation, increase in energy is not equal to $\frac{1}{2}\mathrm{k}\delta^2$.

Hence, option (D) is wrong.

Q.10 Answer:

The speed of the block at the bottom is $7.67\ \mathrm{m/s}$ (if the inclined plane is smooth).

Solution:

Let h be the height of inclined plane

$\Rightarrow \mathrm{h} = 5\sin 37^\circ = 3\ \mathrm{m}$

(a) As the block slides down the inclined plane, it loses GPE and gains KE.

Loss in GPE $=$ gain in KE

$\mathrm{mg}($ **loss in height** $) = \mathrm{KE_f} - \mathrm{KE_i} \Rightarrow \mathrm{mgh} = \frac{1}{2}\mathrm{mv}^2 - 0 \Rightarrow \mathrm{v} = \sqrt{2\mathrm{gh}} = \sqrt{2\times 9.8\times 3} = 7.67\ \mathrm{m/s}$

(b) As the block comes down, it loses GPE.

It gains KE and does work against friction.

Loss in GPE $=$ gain in KE $+$ work done against friction

$\Rightarrow \mathrm{mgh} = \left(1/2\mathrm{mv}^2 - 0\right) + (\mu\mathrm{mg}\cos 37^\circ)\,\mathrm{s}$

$\Rightarrow 3\mathrm{mg} = 1/2\mathrm{mv}^2 + (0.25)\times \mathrm{mg}\times(4/5)\times 5 \Rightarrow \mathrm{v} = \sqrt{4\ \mathrm{g}} = 6.26\ \mathrm{m/s}$

Q.11 Answer:

The work done by normal reaction on triangular block is $W_1 = \frac{-mg}{2}v_t t\sin(2\theta)$

Solution:

(ABCD)

A) As the block a always remains stationary with respect to triangular block $f = mg\sin\theta$

$N = mg\cos\theta$

Displacement of the triangular block w.r.t ground $-v_e t$

Work done by normal reaction $W_1 = -Nv_0 t\cos\left(90^\rho - \theta\right)$

$W_1 = mg\cos\theta t\cos(\pi/2-\theta)$

$= -mg\cos\theta\sin\theta v_c t$

B) If normal reaction is internal force of a system, work done by normal reaction on the system is zero in all frames of reference

@IITJEE_Advanced

$W_1 + W_2 = 0$

C) $W_2 = -W_1 = \frac{mgv_0 t \sin 2\theta}{2}$

D) $W_1 + W_2 = 0$

Q.12 Answer:

$49\ \text{J}$

Solution:

(i) At $\text{x} = 5\ \text{m}, \text{U} = 20 + (5-2)^2 = 29\ \text{J},\ \text{K} = 20\ \text{J}$

Mechanical energy $= \text{E} = \text{U} + \text{K} = 29 + 20 = 49\ \text{J}$

(ii) The greatest or least value of x **will be at that point where** K **is zero and** U **is maximum.**

$\text{U}_{\text{max}} = 49\ \text{J} \Rightarrow 20 + (\text{x}-2)^2 = 49 \Rightarrow (\text{x}-2)^2 = 29$

$\Rightarrow = x - 2 = \pm\sqrt{29} \Rightarrow x = -3.38\ \text{m}, 7.38\ \text{m}$

(iii) KE **is maximum, where** U **is minimum and**

$U_{\text{min}} = 20J$ at $x = 2m$

$K_{\text{max}} = E - U_{\text{min}} = 49 - 20 = 29J$

Q.13 Answer:

$7.38\ \text{m}, -3.38\ \text{m}$

Solution:

(i) At $\text{x} = 5\ \text{m}, \text{U} = 20 + (5-2)^2 = 29\ \text{J},\ \text{K} = 20\ \text{J}$

Mechanical energy $= \text{E} = \text{U} + \text{K} = 29 + 20 = 49\ \text{J}$

(ii) The greatest or least value of x **will be at that point where** K **is zero and** U **is maximum.**

$\text{U}_{\text{max}} = 49\ \text{J} \Rightarrow 20 + (\text{x}-2)^2 = 49 \Rightarrow (\text{x}-2)^2 = 29$

$\Rightarrow = x - 2 = \pm\sqrt{29} \Rightarrow x = -3.38\ \text{m}, 7.38\ \text{m}$

$\Rightarrow \quad x - 2 = \pm\sqrt{29} \Rightarrow x = -3.38\ \text{m}, 7.38\ \text{m}$

(iii) KE **is maximum, where** U **is minimum and**

$U_{\text{min}} = 20J$ at $x = 2m$

$K_{\text{max}} = E - U_{\text{min}} = 49 - 20 = 29J$

Q.14 Answer:

$29\ \text{J}, 2\ \text{m}$

Solution:

(i) At $\text{x} = 5\ \text{m}, \text{U} = 20 + (5-2)^2 = 29\ \text{J},\ \text{K} = 20\ \text{J}$

Mechanical energy $= \text{E} = \text{U} + \text{K} = 29 + 20 = 49\ \text{J}$

(ii) The greatest or least value of x **will be at that point where** K **is zero and** U **is maximum.**

$\text{U}_{\text{max}} = 49\ \text{J} \Rightarrow 20 + (\text{x}-2)^2 = 49 \Rightarrow (\text{x}-2)^2 = 29$

$\Rightarrow = x - 2 = \pm\sqrt{29} \Rightarrow x = -3.38\ \text{m}, 7.38\ \text{m}$

$\Rightarrow \quad x - 2 = \pm\sqrt{29} \Rightarrow x = -3.38\ \text{m}, 7.38\ \text{m}$

(iii) KE **is maximum, where** U **is minimum and**

@IITJEE_Advanced

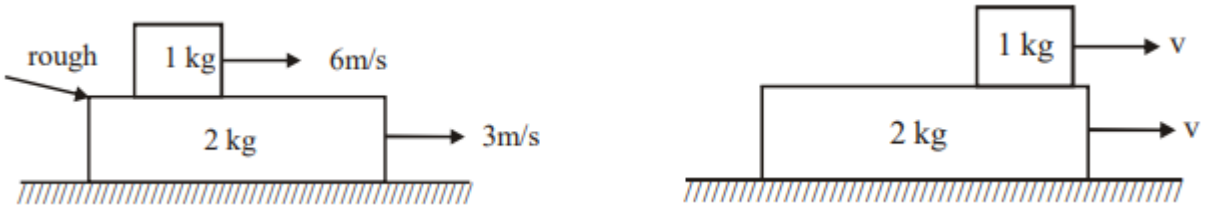
$U_{\min} = 20J$ at $x = 2m$

$K_{\max} = E - U_{\min} = 49 - 20 = 29J$

Q.15 Answer:

Both (A) and (B) are correct

Solution:



When relative motion stop, both block will move with same velocity v, momentum conservation gives $1 \times 6 + 2 \times 3 = 3\mathrm{v} \Rightarrow \mathrm{v} = 4\ \mathrm{m/s}$

Work done by friction on block = change in K.E.

$\Rightarrow \quad \mathrm{W_{f1}} = \frac{1}{2} \times 1 \times (4)^2 - \frac{1}{2} \times 1 \times (6)^2 = -10\ \mathrm{J}$

$\mathrm{W_{f2}} = \frac{1}{2} \times 2 \times (4)^2 - \frac{1}{2} \times 2 \times (3)^2 = 7\ \mathrm{J}$

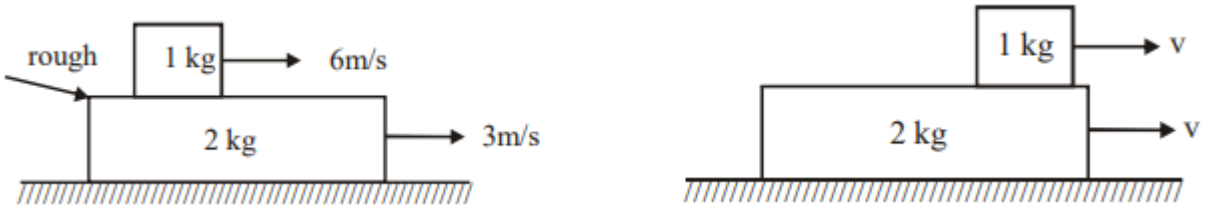
Now total work done by friction $= (-10 + 7) = -3\ \mathrm{J} (= -$ **total loss in K.E.** $)$

$\therefore$ **Total loss in K.E.** $= 3\ \mathrm{J}$

Q.16 Answer:

$-3\ \mathrm{J}$

Solution:



When relative motion stop, both block will move with same velocity v, momentum conservation gives $1 \times 6 + 2 \times 3 = 3\mathrm{v} \Rightarrow \mathrm{v} = 4\ \mathrm{m/s}$

Work done by friction on block = change in K.E.

$\Rightarrow \quad \mathrm{W_{f1}} = \frac{1}{2} \times 1 \times (4)^2 - \frac{1}{2} \times 1 \times (6)^2 = -10\ \mathrm{J}$

$\mathrm{W_{f2}} = \frac{1}{2} \times 2 \times (4)^2 - \frac{1}{2} \times 2 \times (3)^2 = 7\ \mathrm{J}$

Now total work done by friction $= (-10 + 7) = -3\ \mathrm{J} (= -$ **total loss in K.E.** $)$

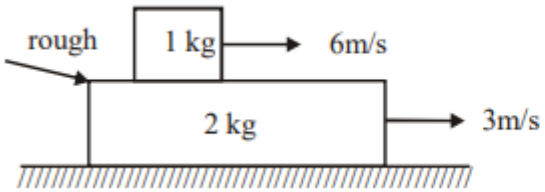
$\therefore$ **Total loss in K.E.** $= 3\ \mathrm{J}$

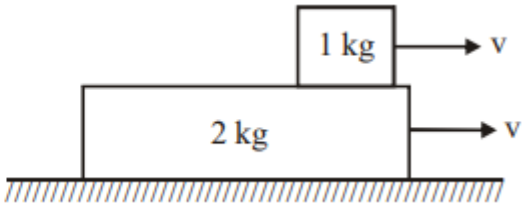
Q.17 Answer:

$3\ \mathrm{J}$

Solution:

@IITJEE_Advanced





When relative motion stop, both block will move with same velocity v, momentum conservation gives $1 \times 6 + 2 \times 3 = 3\mathrm{v} \Rightarrow \mathrm{v} = 4\ \mathrm{m/s}$

Work done by friction on block = change in K.E.

$\Rightarrow \quad \mathrm{W_{f1}} = \frac{1}{2} \times 1 \times (4)^2 - \frac{1}{2} \times 1 \times (6)^2 = -10\ \mathrm{J}$

$\mathrm{W_{f2}} = \frac{1}{2} \times 2 \times (4)^2 - \frac{1}{2} \times 2 \times (3)^2 = 7\ \mathrm{J}$

Now total work done by friction $= (-10 + 7) = -3\ \mathrm{J} (= -$ **total loss in K.E.** $)$

$\therefore$ **Total loss in K.E.** $= 3\ \mathrm{J}$

Q.18 Answer:

A-(q) B-(p) C-(r) D-(p)

Solution:

$(\mathrm{A}) - \mathrm{q}, (\mathrm{B}) - \mathrm{p}, (\mathrm{C}) - \mathrm{r}, (\mathrm{D}) - \mathrm{p}.$

$\mathrm{F}\cos 45^\circ = \mu \mathrm{mg} + \mu \mathrm{F} \sin 45^\circ$

$\mathrm{F} = 75\sqrt{2}$

Surface is exerting only friction.

Work done by friction $= -75\sqrt{2} \times \frac{1}{\sqrt{2}} \times 20 = -1500\ \mathrm{J}$

Work done by gravity is zero as it is perpendicular to the direction of motion.

Work done by man on the block in pushing it through $10\ \mathrm{m}$**,**

$\mathrm{w} = 75\sqrt{2} \times \frac{1}{\sqrt{2}} \times 10 = 750\ \mathrm{J}$

As the block is moving with constant velocity, net force on the block must be zero.

Q.19 Answer:

—

Solution:

$W = \int f_r dx = -k \ln\left(\frac{2}{0.1}\right) = -1.5\ \mathrm{J}$

$\therefore W = \Delta K$

$\frac{1}{2} \times 1 \times v^2 - \frac{1}{2} \times 1 \times 4 = -1.5$

$\Rightarrow v = 1\ \mathrm{m\ s^{-1}}$

Q.20 Answer:

—

Solution:

@IITJEE_Advanced

Let v be the velocity of the particle at the bottom of the circular track.

From the principle of conservation of mechanical energy

We have, $mgh = \frac{1}{2}mv^2$ **or,** $h = \frac{v^2}{2g}$

To complete the vertical loop, $v_{\min} = \sqrt{5gR}$

$\therefore \quad (h)\min = \frac{5gR}{2g} = \frac{5\mathrm{R}}{2}$

@IITJEE_Advanced